

----- Active space -----

HEMATOLOGICAL MALIGNANCIES

Leukemia

00:00:45

- m/c hematological malignancy

Characteristic features :

- Clonal proliferation + arrest of maturation (Tumor cells composed of immature cell series).

Types of leukemia :

- Acute (m/c) >> Chronic
 - ALL (Acute lymphoblastic leukemia).
 - AML (Acute myeloblastic leukemia).

ALL :

Incidence :

- m/c malignancy in children.
- Age group : 2-5yrs.
- Gender : males > females.

Conditions with increased risk for ALL :

- Ataxia telangiectasia
- Bloom's syndrome.
- Severe combined immunodeficiency (SCID).
- Down's syndrome.
- Environmental factors (Eg : Ionizing radiation).
- Fanconi's anemia.
- Le Fraumeni syndrome.

khanirfan0392@gmail.com

Classification of ALL :

1. Immunophenotypic classification (Based on cell type) :

	CD markers (Via flow cytometry)	management
Pre B cell (m/c : 85%)	CD10 (markers of good prognosis), CD19, CD20, & CD45	Good prognosis with treatment
mature B cell (and most common)	CD19, CD20, CD21 S. Ig+ (Surface Ig positivity)	Intensified treatment required
T cell	CD2, CD3, CD4, CD5, CD7, CD8	mediastinal involvement noted + CNS manifestations

2. Cytogenetic classification :

- Based on chromosome number, translocation.

General Features :

----- Active space -----

Bone marrow failure :

- Pancytopenia
 - Anemia
 - Thrombocytopenia (Present as petechiae)
 - Leukopenia
- Exception : Leukocytes may be seen in a few patients.
- Night bone pains (D/t infiltration).

Extramedullary involvement (> 60% cases) :

1. Infiltration of liver and spleen
 - Hepatosplenomegaly
 - Splenomegaly
2. Superficial lymphadenopathy : Non tender, firm, fixed, confluent.
3. Mediastinal involvement :
 - Tracheal compression : unexplained cough/breathing difficulty.
 - SVC syndrome : Compression of SVC → ↓ venous stasis
 - Facial flushing/edema
 - Jugular vein distension
4. CNS leukemia :
 - Cranial neuropathies.
 - Seizures.
 - ↑ ICT features.
5. Testicular leukemia :
 - Painless unilateral/bilateral testicular swelling.

khanirfan0392@gmail.com

Diagnosis :**Bone marrow (Aspiration & Biopsy) :**

- Aspiration may be normal.
- ≥ 25% lymphoblasts : Diagnosis of ALL.

Classification based on prognosis :

	Standard risk/good prognosis	High risk/poor prognosis
Age	2-10 years	≤ 1yr and > 10 yrs
Gender	Female	males
CNS involvement	-	+
Initial WBC count	< 50,000/mm ³	> 50,000/mm ³
Subtypes	Pre-B cell	B cell, T cell
Cytogenetics : To determine prognosis	• Hyperdiploidy (> 50 chromosomes). • Trisomy 4, 10.	• Hypodiploidy (<44 chromosome). • Translocation : - t(9;22) : Ph (Philadelphia chromosome). - t(4;11) : Infantile leukemia (<1 yr old).

216

Childhood Malignancies

----- Active space -----

Treatment :

	Phase	Goal	Medications
I	Induction phase	To induce remission Look for response <pre> graph TD A[To induce remission Look for response] --> B[Good response] A --> C[Minimal Residual Disease (MRD)] B --> D[Good prognosis] C --> E[Good prognosis] </pre> Good response → Good prognosis • ↓ in WBC count • No blast cells in bone marrow & circulation Minimal Residual Disease (MRD) • Cannot be detected by conventional methods. • Test : Flow cytometry/PCR • No MRD → Good prognosis.	- Prednisolone - L-asparaginase - Anthracycline - Vincristine
II	Prophylactic CNS therapy	To treat subclinical CNS involvement	Intrathecal methotrexate
III	Consolidation (Intensification) phase	Tackle drug resistance & sustain remission	- methotrexate - Etoposide - Cyclophosphamide - Cytosine-arabioside
IV	Maintenance phase Once a week for 2 yr	To prevent relapse	- 6-mercaptopurine - methotrexate

AML :

- and m/c type of leukemia

Classification :

I. FAB classification :

M0 : minimal differentiation.

M1 : Without maturation.

M2 : With maturation.

M3 : APML (Acute Promyelocytic leukemia).

II. WHO classification :

M4 : myelomonocytic leukemia.

M5 : monocytic leukemia.

M6 : Erythroleukemia.

M7 : megakaryoblast leukemia.

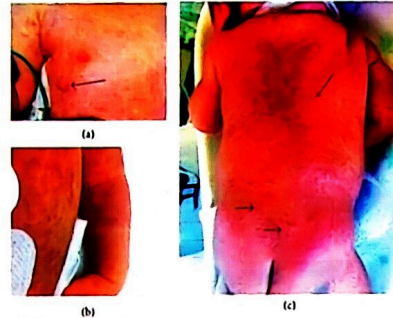
	Favorable risk factors	High risk factors
Genetics	• Inversion 16. • t(8; 21). • CEBPA gene mutation (Codes for transcription factor for maturation of myeloid cells). • NPM1 gene mutation.	• monosomy 5/7. • 5q deletion.
Response to therapy	• Chemotherapy.	• Hematopoietic Stem Cell Transplantation (HSCT).

Clinical Features :

1. Similar to ALL.

Exception : Hepatosplenomegaly rarely seen (Present in **m4, m5 variety**), No lymphadenopathy.2. **Very high WBC count** → **Leukostasis** → Lung infiltrates
→ Stroke manifestations

3. Disseminated Intravascular Coagulation (DIC) in m3 varieties.



4. Gum Hypertrophy (Seen in m4 subtype)

Diagnosis :Bone marrow aspiration : **>20% myeloblasts.****Treatment :**

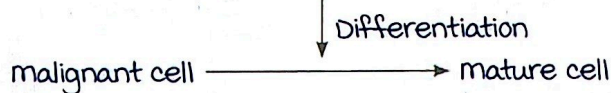
1. Induction phase :

- Cytosine arabinoside.
- Daunorubicin.

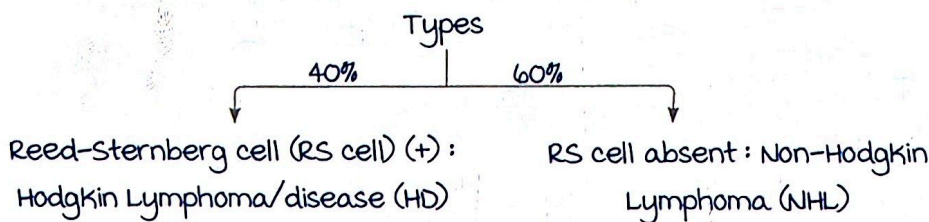
2. Consolidation phase :

- Cytosine arabinoside.
- Etoposide.

Drugs used in m3 (APML) : ATRA (All trans retinoic acid)/tretinoin

5. **Chloroma/myeloid sarcoma** : Deposition of leukemia cells in skin (D/t extramedullary hematopoiesis)**Lymphoma**

00:33:47

**HODGKIN'S LYMPHOMA (HL) :**

Age group : 15-19 years.

Gender : male predominance.

Risk factors → Infections: HHV6, CMV, **EBV** (↑ risk by 4X)

→ Autoimmune disorders : SLE

Feedback

----- Active space -----

Clinical features :**1. Lymphadenopathy (m/c : Cervical) :**

- Painless.
- matted.
- Firm rubbery consistency.

2. Constitutional symptoms :

- High-grade fever ($> 38^{\circ}\text{C}$).
- Drenching night sweats.
- **Weight loss** ($> 10\%$ from last 6 months).

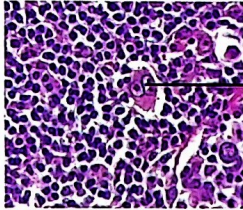
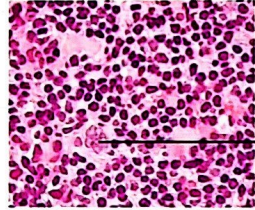
If absent

Grade A (Good prognosis)

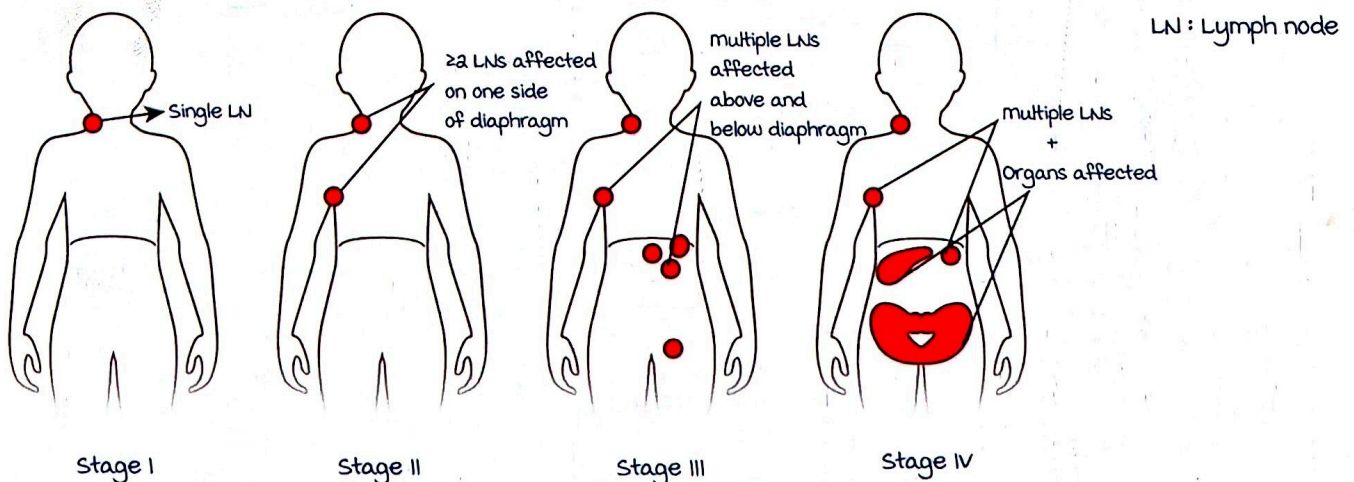
If present

Grade B (Poor prognosis)

Types :

	Classical	Nodular-lymphocyte predominant
Positive CD markers	CD15, CD30	CD20, CD45
RS cell on excisional biopsy of lymph node (malignant transformation of germinal B cell)	 Owl eye inclusion body	 Popcorn cell

Subtypes of Classical HD	Epidemiology	RS cell variant	EBV association	Prognosis
1. Nodular sclerosis	m/c worldwide : F > M	Lacunar	-	Excellent
2. mixed cellularity	m/c in India : M > F	-	+	Good
3. Lymphocyte rich	-	mononuclear		Excellent
4. Lymphocyte depleted	-	Reticular		Poor

Treatment :Based on the **modified Ann Arbor Staging :**

Stage or Grade	I		II		III		IV		Any stage with	
	IA	IB	IIA	IIB	IIIA	IIIB	IVA	IVB	X	E
Clinical presentation	Favorable					Unfavorable				
Treatment	• Chemotherapy : ABVD . A : Adriamycin B : Bleomycin V : Vinblastine D : Dacarbazine					• Chemotherapy + • Radiotherapy + Involved Field Radiation Therapy (IFRT)				

----- Active space -----

- X : Bulky > 10 cm size.
- E : Extension into extralymphatic site.

NON-HODGKIN'S LYMPHOMA (NHL) :

- Age group : 5 - 15 years.

Types of NHL :

Based on lymphocyte of origin

B cell type

- Burkitt's lymphoma (m/c worldwide).
- Diffuse large B cell lymphoma (DLBCL).

T cell type

- Lymphoblastic lymphoma (m/c in India).
- Anaplastic large cell lymphoma.

Intermediate cell

Clinical features :**B cell type (Burkitt Lymphoma) :**

	Endemic	Sporadic
m/c	Worldwide (Endemic to africa)	In India
Presentation :	 1. Jaw lesions - Lytic lesions of bones. - Draining sinuses. - Diffuse swelling. 2. Generalized abdominal distension (D/t intra-abdominal tumor)	• Intra-abdominal tumors. • O/E : - mass P/A. - unexplained abdominal pain. - Intussusception (with tumors as lead point).
Associated with EBV	(+)	↓ ↓

T cell type :

- Similar to ALL.
- Presents as mediastinal mass.

----- Active space -----

St. Jude Staging :

	Lymph Node involvement	Other lesions
Low risk {	Stage 1	Single
	Stage 2	None
High risk {	Stage 3	Single resectable GI lesion.
	Stage 4	Non-resectable GI lesions
		Non resectable GI lesions + bone marrow/CNS involvement

Treatment :

- malignant chemotherapy :
 1. Cyclophosphamide.
 2. Vincristine.
 3. Prednisolone.
 4. Doxorubicin.
 - (+)
 5. methotrexate if High risk.
- } Low risk

Langerhans Cell Histiocytosis (LCH)

00:58:41

- AKA Histiocytosis X.
- Clonal proliferation of Langerhans like cells.

Clinical Features :

- Wide spectrum of disease (Can be multisystemic).

Sites involved :

1. **Bone** : m/c.
 - Skull > jaw (missing/loose teeth).
2. **Skin** :
 - Scales (+) discharge.
 - mimics seborrheic dermatitis and does not respond to Rx of the same.
3. **Pituitary infiltration.**

Stalk destruction → Block in release of vasopressin/ADH → Diabetes Insipidus (DI)
(From posterior pituitary granules)

↓

mRI : Loss of bright spot in posterior pituitary
4. Lymphadenopathy.
5. Hepatosplenomegaly.



Lytic lesions in skull



Cradle cap

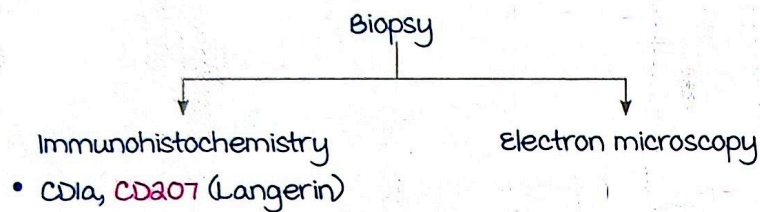
LCH Clinical Spectrum :

Spectra of LCH	Age group	Clinical features
Eosinophilic granuloma	Older children	Lytic bone in skull
Hand-Schüller-Christian disease	2-6 years	Skull defect Exophthalmos
Letterer-Siwe disease (Severe form)	< 2 years	- Seborrheic dermatitis. - Ear discharge. - System involvement : - Hepatosplenomegaly (Abnormal LFT). - Lymphadenopathy.

----- Active space -----

Increasing severity

Diagnosis :



Tennis racket appearance

Birbeck granules

Treatment :

1. Simple lesion :

- Curettage.
- Intralesional steroids.

2. Multiple lesion :

- Vinblastine (m/c).
- 6 MP (mercaptopurine).
- Prednisolone.

3. Refractory disease :

- Cladribine.